Secure wiring and finish the design

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Document permanent connections and evaluate added mass or friction.

SUCCESS CRITERIA - I can...

1. Check circuit continuity and mechanical security.
2. Preserve moving-part clearance during decoration.
3. Relate a design revision to an energy transfer or force.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “strain relief” and “insulation” to explain your model or result. Sentence starters are optional.

strain relief: support that keeps a pull from stressing an electrical joint

insulation: material that limits unwanted electric conduction

short circuit: an unintended low-resistance path bypassing a component

01 Bring back an earlier idea

A 40-tooth driver turns a 240-tooth gear.
Give ideal output/input torque and speed ratios and state the loss assumption.

Word help: torque = the turning effect of a force about an axis

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: I = 0 with power disconnected.
Disconnect before soldering/securing wires. Use insulation and strain relief; route wires clear of gears. Follow supervised tool procedures.

Worked reasoning: A secure circuit also needs strain relief: a pull on a wire should not be carried solely by a solder joint. Insulation prevents exposed conductors from making unintended contacts.

Your turn: Draw one wire route showing a connection, insulation, strain relief and clearance from a rotating axle.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: ① ② ③

03 Represent and interpret

Draw your final wiring route.
Mark soldered joints, insulation, strain relief and the switch location.
Identify one possible wheel or gear interference point.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: ① ② ③

04 Solve and justify

A supplied car has net force .30 N.
Compare acceleration at .20 kg and .25 kg.
Find the percent reduction under the fixed-force assumption.

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

Secure wiring and finish the design

antocciphysics

Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Analyze the reasoning

A wire rubs a wheel.
Explain the possible effect on energy transfer and propose a repair before another powered test.

Sentence starter: The step I would change is … . The evidence is … .

Marzano check - circle: 1 2 3

06 Choose a method

Near an ideal long straight wire, current becomes 4 times as large and distance becomes 6 times as large.
Compare B.

Word help: current = rate of charge flow

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3

07 Exit check

Explain why “lighter is always faster” is too strong for a powered car.
Name two other influential factors.

Sentence starter: Under the assumption …, the evidence … supports … because … .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A wire fault adds 2 ohms in series to a supplied motor model of 4 ohms on 6 V.
Compare current before/after and power in the unwanted resistance.
Explain why a warm joint deserves investigation.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3